Monte Carlo Path Tracing Renderer

From the Rendering Equation to Physically Based Rendering · COMP4610/8610 Research Project 2

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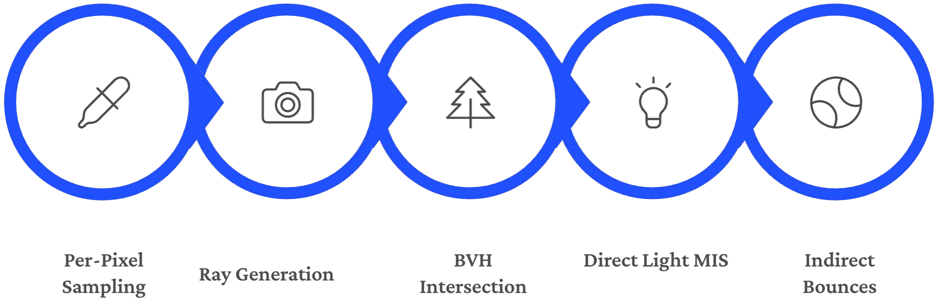
Team Collision

Rendering Pipeline Overview



6 BSDF Materials

Diffuse · GGX Microfacet · Metal · Mirror · Glass · Emit

BVH Acceleration Structure

~28 ms/ray for a 400k-triangle scene

Multiple Importance Sampling

Power Heuristic β=2, dramatically reduces variance

C++17 · OpenMP multithreading · ~4000 lines of code

Material System — BSDF Architecture & GGX Microfacet

DIFFUSE

Lambertian, cosine-weighted hemisphere sampling

MICROFACET

Cook-Torrance, GGX normal distribution function

METAL

GGX highlights, color = F_0 term

MIRROR

Perfect specular reflection (Delta BSDF)

GLASS

Fresnel refraction + reflection

EMIT

Self-emissive surface, acts as area light

GGX Core Formula

$$f_r = (1 - m)\frac{\text{albedo}}{\pi} + m\frac{D \quad G \quad F}{4 \mid n \quad \omega_i \mid \mid n \quad \omega}$$

D — GGX Normal Distribution

\alpha = \text{roughness}^2, concentrated in the specular lobe

G — Smith-Schlick Geometry Shadowing

Jointly models self-shadowing and mutual occlusion

F — Schlick Fresnel Approximation

Grazing angles increase highlight intensity

Material factory auto-registration — adding a new type does not require modifying the central dispatch code

Lighting & Environment Illumination

7 Types of Analytical Light Sources

1 **AreaRect · AreaDisk · AreaSphere**

2 **Point · Spot · Directional · MeshLight**

Importance sampling by luminance CDF · translucent shadows (cumulative colored transmittance) · delta light sources handled automatically

HDR Environment Map

- `stb_image` loads .hdr lat-long projection
- 2D luminance-weighted CDF importance sampling O(log n)
- Supports rotation + intensity scaling
- MIS combined with BSDF sampling significantly reduces noise

After environment maps are enabled, shadows go from pitch black → natural soft light, and MIS joint sampling converges much faster

Results — Rendered Images

Desktop Typewriter Scene · 1200×800 · 64 spp · 8 bounces · Adaptive + Denoise

Front View

GGX metal highlights · soft contact shadows

Side View

Side lighting · material performance across lighting angles

Translucent Lampshade

Thin translucent material · warm backlighting · texture/normal detail

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Retro Room Scene · 1200×600 · 64 spp · envmap + window area light · Denoise

Essential

Core desk and chairs · indirect diffuse global illumination

Grouped

Full room · multiple environment light bounces

Denoised

Joint bilateral filter · preserves highlight detail and shadow depth

Limitations & Future Work

Current Limitations

No Depth of Field

Aperture/focus fields are reserved; thin-lens sampling is not implemented

Naive Median-Split BVH

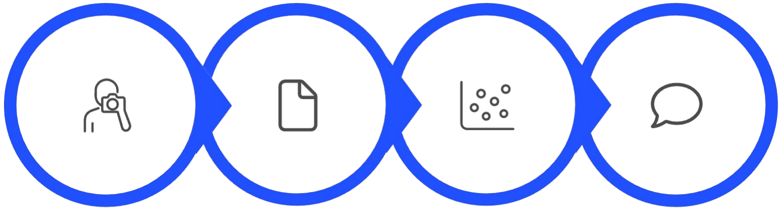
Introducing SAH can improve traversal efficiency by 20–40%

No Volumetric Scattering

Does not support participating media rendering such as fog or smoke

Implemented: 6 BSDFs · 7 light types · HDR environment lighting · MIS · BVH · adaptive sampling · denoising · translucent shadows · complex scenes with 400K triangles

Future Directions



Thin-Lens Depth of Field

SAH BVH Optimization

Volumetric Scattering

GPU Acceleration

Thank You — Q&A